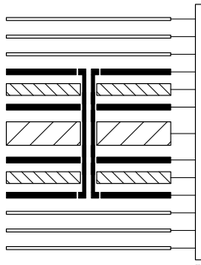
# Main board Fabrication Document

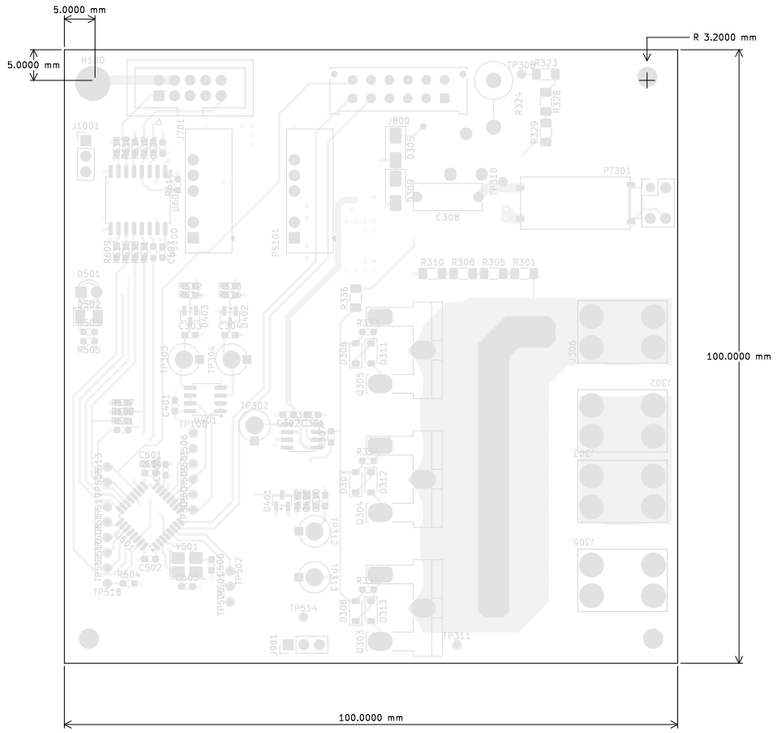
## Top Fabrication (Scale 1:1)

## Layer Stack Legend



| Material | Layer | Thickness | Dielectric | Type | Gerber |
|---|---|---|---|---|---|
| | F.Paste | | | Paste Mask | |
| | F.Silkscreen | | Direct Printing | Legend | GBR |
| | F.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| **Copper** | **F.Cu (Sig, PWR)** | **0.07mm (2.00oz)** | | **Signal** | **GBR** |
| Prepreg | | 0.1mm | FR4_7628 | Dielectric | |
| **Copper** | **In1.Cu (GND)** | **0.035mm (1.00oz)** | | **Plane** | **GBR** |
| Core | | 1.828mm | FR4 | Dielectric | |
| **Copper** | **In2.Cu (Sig, PWR)** | **0.035mm (1.00oz)** | | **Plane** | **GBR** |
| Prepreg | | 0.1mm | FR4_7628 | Dielectric | |
| **Copper** | **B.Cu (Sig, PWR)** | **0.07mm (2.00oz)** | | **Signal** | **GBR** |
| | B.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| | B.Silkscreen | | Direct Printing | Legend | GBR |
| | B.Paste | | | Paste Mask | |

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating




FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL SNPB

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

| | |
|---|---|
| BOARD SIZE | 100.000 × 100.000 mm |
| BOARD THICKNESS | 2.278 mm |
| TRACE WIDTH | 0.200 mm |
| TRACE TO TRACE | 0.200 mm |
| MIN. HOLE (PTH) | 0.300 mm |
| MIN. HOLE (NPTH) | 1.000 mm |
| ANNULAR RING | 0.150 mm |
| COPPER TO HOLE | 0.300 mm |
| COPPER TO EDGE | 0.500 mm |
| HOLE TO HOLE | 0.250 mm |

Lim. resistor
PFC

HMI

Ext
ctrl

SWD
Status

Lamp
trig

1 kV strobe
04/Jul/2025

Boot

Cap+

Lamp+

Light
sense

Felipe L Rodrigues

Cap−

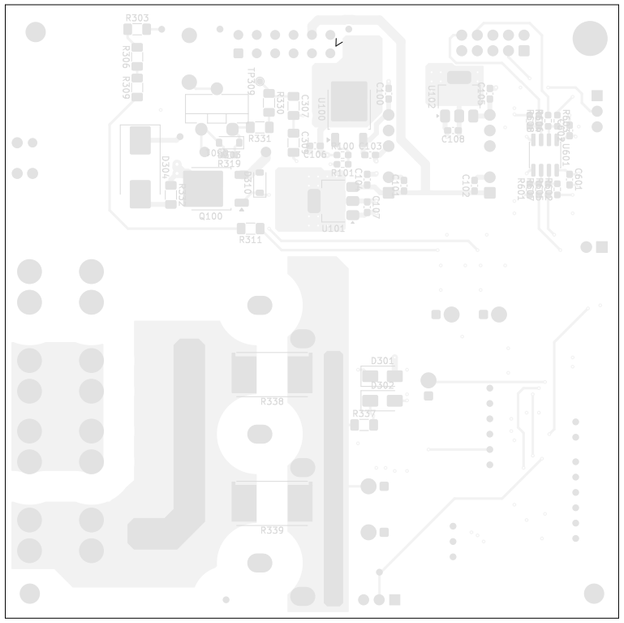
Lamp−

ChurrasCod
Electronics

| | | | | |
|---|---|---|---|---|
| Comments: | Company: ChurrasCod Electronics | | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | Board Name: **Main board** | | Project Name: **1 kV strobe** | |
| Sheet Title: Top Fabrication (Scale 1:1) | File Name: Strobe_main.kicad_pcb | Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: | Size: **A3** | Sheet: **1** of **10** |

# Main board Fabrication Document

**Bottom Fabrication (Scale 1:1)**




1 kV strobe
04/Jul/25
Felipe L Rodrigues

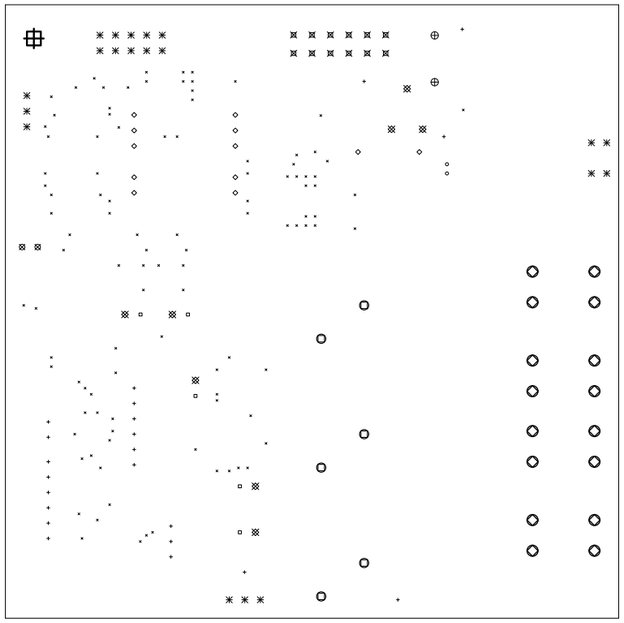
| | | | | |
|---|---|---|---|---|
| ChurrasCod Electronics | Comments: | Company: **ChurrasCod Electronics** | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | | Board Name: **Main board** | Project Name: **1 kV strobe** | |
| | Sheet Title: Bottom Fabrication (Scale 1:1) | File Name: Strobe_main.kicad_pcb / Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| | Sheet Path: | Reviewer: | Size: **A3** | Sheet: **2** of **10** |

# Main board Fabrication Document

## Drill Drawing L1 - L4 (Scale 1:1)

### Drill Table

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 105 | 0.30mm (11.81mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Via |
| ○ | 2 | 0.50mm (19.69mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Via |
| + | 22 | 0.50mm (19.69mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| □ | 5 | 0.70mm (27.56mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ◇ | 12 | 0.80mm (31.50mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ⊠ | 2 | 0.90mm (35.43mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ✳ | 20 | 1.00mm (39.37mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ⊠ | 12 | 1.02mm (40.16mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ⊠ | 8 | 1.10mm (43.31mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ⊕ | 2 | 1.20mm (47.24mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ○ | 6 | 1.50mm (59.06mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ○ | 16 | 1.80mm (70.87mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| ⊕ | 1 | 3.20mm (125.98mils) | PTH | Round | F.Cu (Slg, PWR) - B.Cu (Slg, PWR) | Pad |
| | Total 213 | | | | | |



ChurrasCod Electronics

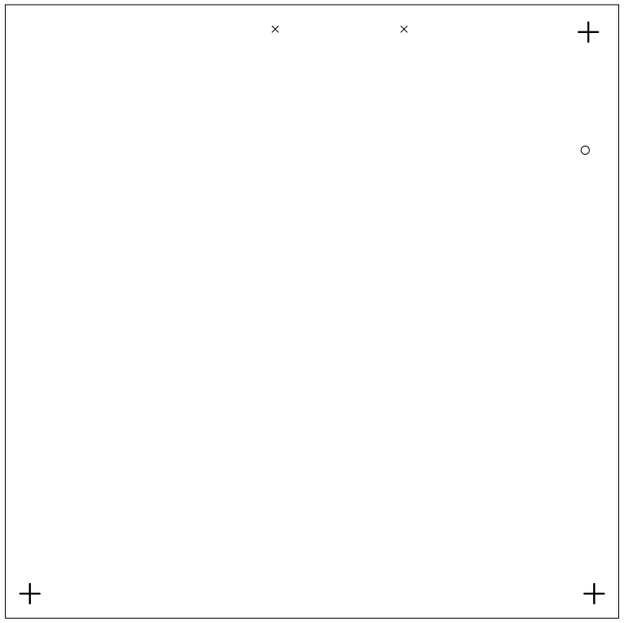
| | | | | |
|---|---|---|---|---|
| Comments: | Company: ChurrasCod Electronics | | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | Board Name: Main board | | Project Name: 1 kV strobe | |
| Sheet Title: Drill Drawing (L1 - L4) | File Name: Strobe_main.kicad_pcb | Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A3 | Sheet: 3 of 10 |

# Main board Fabrication Document

## Drill Drawing L1 - L4 (Scale 1:1)

### Drill Table

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 2 | 1.02mm (40.16mils) | NPTH | Round | F.Cu (Sig, PWR) - B.Cu (Sig, PWR) | Mechanical |
| ○ | 1 | 1.35mm (53.15mils) | NPTH | Round | F.Cu (Sig, PWR) - B.Cu (Sig, PWR) | Mechanical |
| + | 3 | 3.20mm (125.98mils) | NPTH | Round | F.Cu (Sig, PWR) - B.Cu (Sig, PWR) | Mechanical |
| | Total 6 | | | | | |

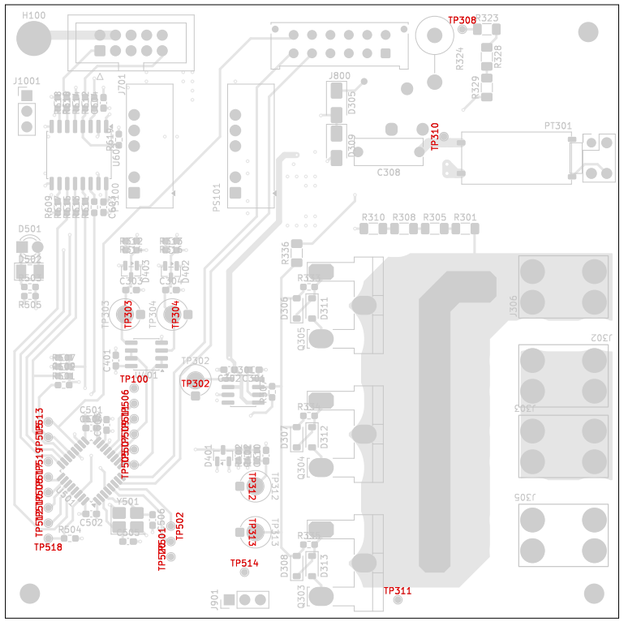


ChurrasCod Electronics

| | | | | |
|---|---|---|---|---|
| Comments: | Company: ChurrasCod Electronics | | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | Board Name: Main board | | Project Name: 1 kV strobe | |
| Sheet Title: Drill Drawing (L1 - L4) | File Name: Strobe_main.kicad_pcb | Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A3 | Sheet: 4 of 10 |

# Main board Fabrication Document

## Top Test Points (Scale 1:1)




| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|
| TP100 | SH0 | 26.00 | 42.50 |
| TP302 | LMP_EN | 36.00 | 42.50 |
| TP303 | SH0 | 25.77 | 54.51 |
| TP304 | Mon_S1k | 33.50 | 54.50 |
| TP308 | S1k | 79.50 | 101.00 |
| TP310 | TP_PRI | 76.50 | 83.50 |
| TP311 | LMP | 69.00 | 8.00 |
| TP312 | SH0 | 44.50 | 26.50 |
| TP313 | SH0 | 44.50 | 19.00 |
| TP501 | Net-(U501-PA0) | 32.00 | 17.50 |
| TP502 | Net-(U501-PA1) | 32.00 | 20.00 |
| TP505 | Mon_S1k_buf | 26.00 | 30.00 |
| TP506 | Net-(U501-PB0) | 26.00 | 40.00 |
| TP507 | Mon_IL_prot | 26.00 | 32.50 |
| TP508 | Net-(U501-PB3) | 12.00 | 25.50 |
| TP509 | Mon_LMP_buf | 26.00 | 35.00 |
| TP510 | Net-(U501-PB4) | 12.00 | 23.00 |
| TP511 | Net-(U501-PA7) | 26.00 | 37.50 |
| TP512 | Net-(U501-PB5) | 12.00 | 20.50 |
| TP513 | Vbus_ctrl | 12.00 | 37.00 |
| TP514 | Light_sns | 44.00 | 12.50 |
| TP515 | INT_EN | 12.00 | 34.50 |
| TP517 | Net-(U501-PA15) | 12.00 | 28.00 |
| TP518 | BOOT0 | 12.00 | 18.00 |
| TP519 | SH0 | 12.00 | 30.50 |
| TP520 | SH0 | 32.00 | 15.00 |

Lim. resistor
PFC

HMI

Ext
ctrl

SWD
Status

Lamp
trig

1 kV strobe
04/Jul/2025

Cap+

Lamp+

Boot

Light
sense

Felipe L Rodrigues

Cap−

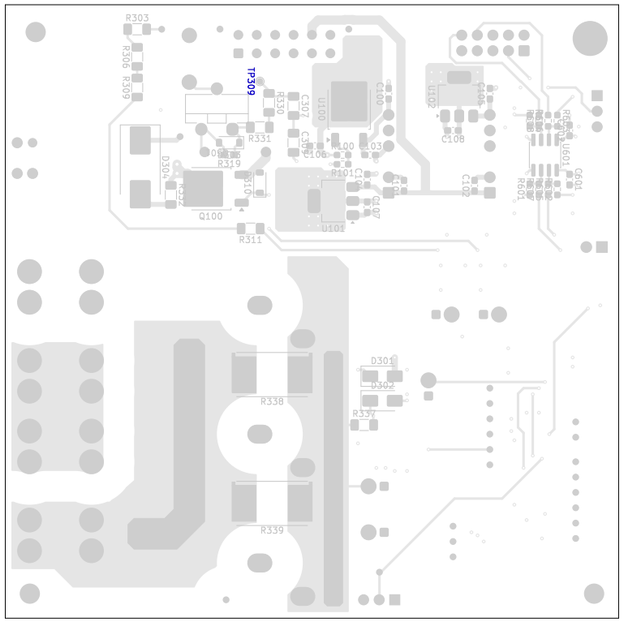
Lamp−

ChurrasCod
Electronics

| | | | | |
|---|---|---|---|---|
| Comments: | Company: ChurrasCod Electronics | | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | Board Name: Main board | | Project Name: 1 kV strobe | |
| Sheet Title: Top Test Points (Scale 1:1) | File Name: Strobe_main.kicad_pcb | Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A3 | Sheet: 5 of 10 |

# Main board Fabrication Document

## Bottom Test Points (Scale 1:1)




| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|
| TP309 | TP_REF | 63.50 | 92.50 |

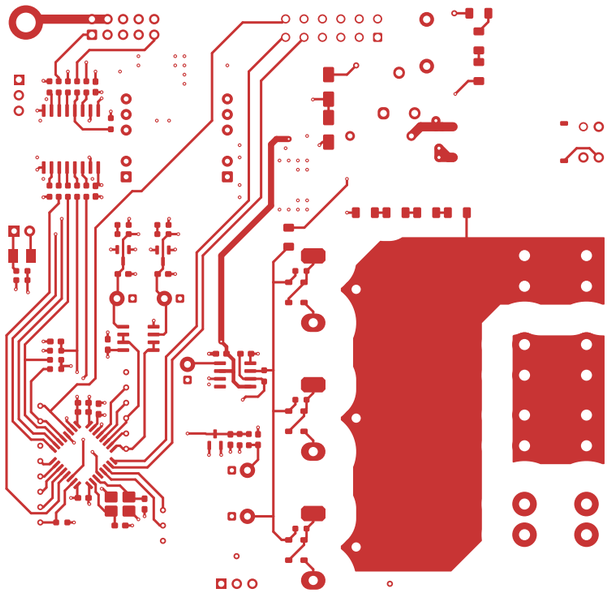
1 kV strobe
04/Jul/25
Felipe L Rodrigues

ChurrasCod Electronics

| | | | | |
|---|---|---|---|---|
| Comments: | Company: ChurrasCod Electronics | | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | Board Name: Main board | | Project Name: 1 kV strobe | |
| Sheet Title: Bottom Test Points (Scale 1:1) | File Name: Strobe_main.kicad_pcb | Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A3 | Sheet: 6 of 10 |

# Main board Fabrication Document

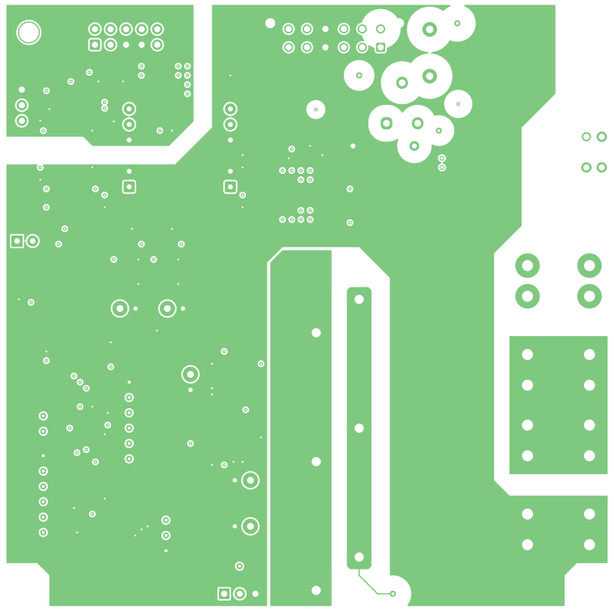
F.Cu (Sig, PWR) (Scale 1:1)



| | | | | |
|---|---|---|---|---|
| ChurrasCod Electronics | Comments: | Company: ChurrasCod Electronics | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | | Board Name: **Main board** | Project Name: **1 kV strobe** | |
| | Sheet Title: F.Cu (Sig, PWR) (Scale 1:1) | File Name: Strobe_main.kicad_pcb / Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| | Sheet Path: | Reviewer: | Size: **A3** | Sheet: **7** of **10** |

# Main board Fabrication Document

In1.Cu (GND) (Scale 1:1)



| | | | | |
|---|---|---|---|---|
| Comments: | Company: ChurrasCod Electronics | | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | Board Name: Main board | | Project Name: 1 kV strobe | |
| Sheet Title: In1.Cu (GND) (Scale 1:1) | File Name: Strobe_main.kicad_pcb | Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A3 | Sheet: 8 of 10 |

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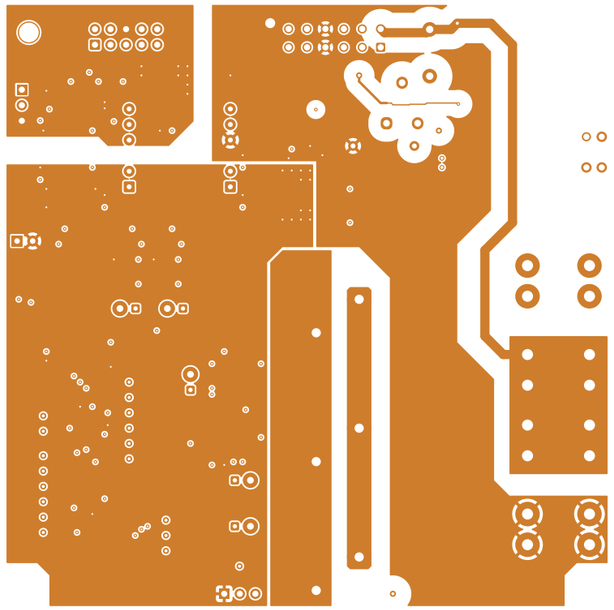
# Main board Fabrication Document

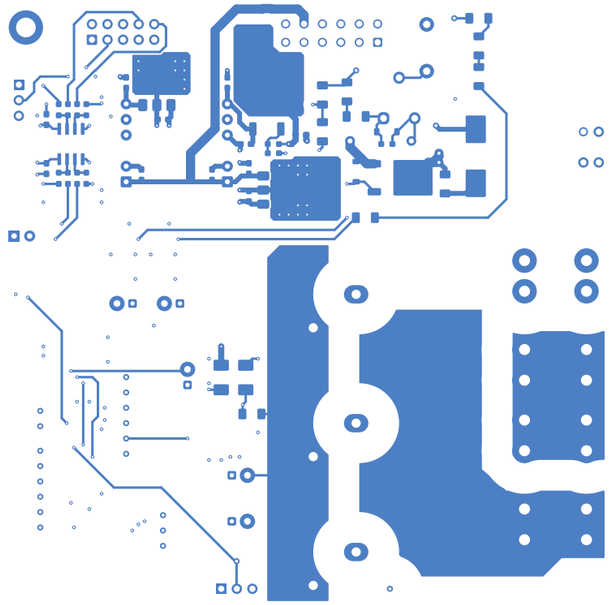
In2.Cu (Sig, PWR) (Scale 1:1)



| | | | | |
|---|---|---|---|---|
| ChurrasCod Electronics | Comments: | Company: ChurrasCod Electronics | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | | Board Name: Main board | Project Name: 1 kV strobe | |
| | Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1) | File Name: Strobe_main.kicad_pcb | Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| | Sheet Path: | | Reviewer: | Size: A3 | Sheet: 9 of 10 |

# Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



| | | | | |
|---|---|---|---|---|
| ChurrasCod Electronics | Comments: | Company: ChurrasCod Electronics | Variant: PRELIMINARY | Git Hash: 4b77bde |
| | | Board Name: Main board | Project Name: 1 kV strobe | |
| | Sheet Title: B.Cu (Sig, PWR) (Scale 1:1) | File Name: Strobe_main.kicad_pcb | Designer: FR | Date: 2024-04-13 | Revision: 1.0.0+ (Unreleased) |
| | Sheet Path: | | Reviewer: | Size: A3 | Sheet: 10 of 10 |